

PAANIPANCHAYAT

Fair Water. Peaceful Farming. — Comprehensive Project Documentation

Platform: AI-Powered Water Sharing & Dispute Mediation Platform for Indian Farmers

1. Executive Summary & Problem Statement

In canal-irrigated agricultural regions across India, water scarcity and unequal distribution cause severe socio-economic distress. Upstream farmers often over-extract water, resulting in tail-end starvation where downstream farmers receive little to no water for their parched crops. Existing distribution systems lack transparency, leading to bitter interpersonal disputes, delayed resolution, and compromised crop yields.

PaaniPanchayat solves this systemic crisis by combining mathematical optimization (Google OR-Tools) with democratic transparency and AI-powered dispute mediation (LangGraph / LangChain). It computes scientific crop-stage water needs, allocates available canal water with verified fairness, and provides automated dispute resolution with multi-lingual support in English, Hindi, and Marathi.

2. Core Platform Capabilities

- **Linear Optimization with Google OR-Tools:** Solves canal water distribution subject to physical constraints (total volume $\leq$ supply, non-negativity) while optimizing for maximum equity.
- **Scientific Crop Hydrology (FAO-56):** Calculates requirements dynamically using acreage, crop type (\$), growth stage, soil type, irrigation efficiency, and weather credits.
- **3-Day Policy Cooldown Protection:** Prevents quota gaming and hoarding by restricting farmers to one active water request per 3-day cycle.
- **AI Panchayat Dispute Mediation:** Evaluates farmer objections against hydrology facts, generating win-win compromise proposals and immutable audit trails.
- **Trilingual Localized UX:** Fully localized user interfaces in English, Hindi (■■■■■), and Marathi (■■■■■).
- **Progressive Web App (PWA):** Mobile-first, installable directly onto farmers smartphones with offline cache capability.

3. System Architecture Overview

PAANIPANCHAYAT — SYSTEM ARCHITECTURE

AI-Powered Water Sharing & Dispute Mediation Platform | Google OR-Tools & LangGraph

1. PRESENTATION LAYER (Client & Edge PWA)

Next.js 16 + React 19 + TypeScript + Tailwind CSS | Multi-lingual (EN, HI, MR)

CLIENT TIER

- **Farmer Self-Service Dashboard**
Crop stage inputs, requirement explanations, fairness rating, irrigation schedules & objection filing
- **Panchayat Administration Matrix**
Canal supply vs demand balance, hard-constraint status, batch farm reallocation, locked audit records
- **Farmer Profile & Security Controls**
Verified shareholder identity, token-based session popover, contact desk, dispute escalation helpline

2. API GATEWAY & BUSINESS ENGINE

FastAPI (Python 3.13) + Pydantic v2 + Bearer Token Security

BACKEND TIER

- **REST Endpoints (/api/allocation, /api/auth, /api/farms)**
CORS-enabled async routing, request validation & rate-limiting policies
- **Water Requirement Engine (FAO-56 Methodology)**
Dynamic calculation: $\text{Area} * \text{Base rate} * \text{Kc} * \text{Soil} / \text{Efficiency} - \text{Credits}$
- **3-Day Policy Cooldown Service**
Prevents quota hogging by enforcing single active water request per 3 days
- **Weather Intelligence Service**
OpenWeatherMap integration deducting forecast rainfall credits dynamically
- **Audit Logging Subsystem**
Cryptographic verification and immutable decision logs for transparency

3. OR-TOOLS OPTIMIZER & AI MEDIATION COR

AI & SOLVER CORE

Google OR-Tools Constraint Programming & LangGraph Mediation Engine

- **Google OR-Tools Linear Optimizer**
Hard Constraints: $\text{Total} \leq \text{Canal Capacity}$, Non-Negativity, Critical stage limits
- **Multi-Objective Fairness Function**
Minimizes unmet farmer demand while maximizing proportional equity score
- **LangGraph AI Dispute Mediator Agent**
Multi-agent LLM analysis parsing objections against canal rules and crop hydrology
- **Compromise Proposal Generator**
Generates fair reallocation compromises with re-validation by the OR-Tools solver
- **Transparent Reasoning Logs**
Generates human-readable explanations in English, Hindi, and Marathi

4. DATA PERSISTENCE & SYSTEM STATE LAYER

SQLite 3 + SQLAlchemy ORM with ACID Transaction Safety

DATA TIER

- **Relational Data Models**
Users (RBAC), WaterSources (Canal Capacity), Farms (Acreage/Soil/Kc), Allocations (Liters/Time Slots)
- **Disputes & Mediation Store**
Objections, proposed reallocations, audit trail, and signed acceptance status logs
- **Seed & Initial Datasets**
Pre-configured canal rotation cycles, crop coefficient lookup tables, and test farmer profiles

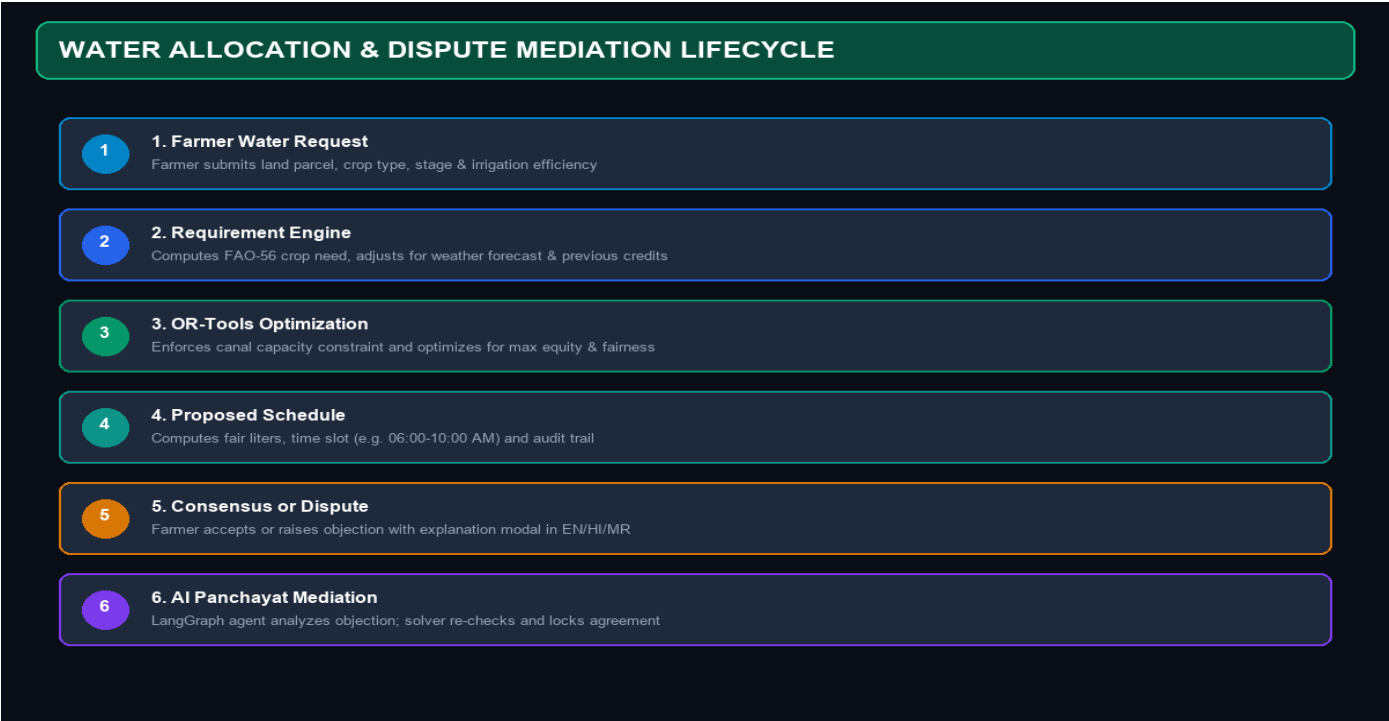
4. Technical Architecture & Technology Stack

Layer	Technologies	Purpose
Presentation Tier	Next.js 16, React 19, TypeScript, Tailwind CSS	Trilingual PWA, Farmer Dashboard, Admin Matrix
API Gateway	FastAPI 0.110+, Python 3.13, Pydantic v2	High-performance async REST API, RBAC Bearer Tokens
Optimization Engine	Google OR-Tools 9.9 (Linear Solver)	Constraint satisfaction and fairness optimization
AI Mediation	LangGraph, LangChain, Google Gemini LLM	Automated dispute resolution and reasoning generation
Database Layer	SQLite 3, SQLAlchemy 2.0 ORM	ACID transactions, farmer records, audit logs
External Services	OpenWeatherMap API	Real-time rainfall forecasting and precipitation credits

5. Optimization & Water Allocation Algorithm

The linear program balances canal supply against aggregate demand while maximizing proportional fairness:

- **Capacity Constraint:** $\sum_{i=1}^N x_i \leq V_{\text{available}}$, where x_i is allocation to farm i .
- **Upper Bound Constraint:** $x_i \leq D_i$, where D_i is calculated demand.
- **Critical Stage Protection:** Crops in critical stages (e.g., Flowering) receive prioritized minimum thresholds.
- **Objective Function:** $\text{Maximize } \sum_{i=1}^N w_i \cdot x_i - \lambda \cdot \text{Variance}$, enforcing high fairness scores (0-100).



6. Security, Privacy & Audit Compliance

- **Zero Data Leakage:** Sensitive environment variables, encryption keys, and database files are isolated via comprehensive .gitignore rules.

- **Role-Based Access Control (RBAC):** Strict role differentiation between Farmers and Panchayat Administrators.
- **Immutable Audit Trail:** Every allocation run, objection submission, and mediation acceptance generates a tamper-evident audit record.